



CAT 6 FLASHER 12 V ENDURANCE TEST

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Part Name	CAT 6 FLASHER 12 V	Application	COMMERCIAL APPLICATION
Customer Part No.	B00 861 544 00 32	Customer	FORCE MOTOR
UCAL Part No.	400230122002	Report No	R&D/CAT6/12VF/DUR/26/030

INTRODUCTION:

UCAL is currently supplying 12V CAT 6 Flashers to Force Motor for commercial applications. This validation plan has been prepared to define the test requirements and validation activities for the current production parts of the respective product. The testing will be carried out as per the approved validation plan to verify product performance, functional reliability, and compliance with specified requirements under defined test conditions.

PURPOSE:

To evaluate the 12 V CAT 6 flashers assembly after Endurance Test .

TEST SPECIFICATION:

Test standard as per customer approved drawing and sign off DVP.

ACCEPTANCE CRITERIA:

Class A: All function of the flasher performs as designed during and after the test.

RESULT:

SAMPLE NO	PERFORMANCE	APPEARANCE	JUDGEMENT
S:01	O	O	O
S:02	O	O	O

O: OK

X: NG

Δ: To be confirmed

CONCLUSION:

The performance of the Flasher assembly is acceptable after test. The samples are met the requirements.

 Mr. Rahul Bongarde	 Mr. Karthikeyan N
Prepared By	Approved By



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PERFROMACE TEST SPECIFICATIONS:

SNO	PARAMETERS	SPECIFICATIONS
1	Operating voltage	10 V~16 V
2	Test voltage	13.5 V
3	Flash rate(Turn signal/Hazard warning)	60~120 Flashes/Minute
4	One lamp failure flashing rate	>140 Flashes/Min(or)1.8 times of flash rate whichever is greater
5	Duty Cycle	40% ~60%

TEST PROCEDURE:

Turn Signal:

- Operating Room Temperature: 25 ± 2 °C
- Testing Voltage: 13.5 V
- Testing Current: Rated Load (Turn Signal)
- Test Duration: 500 H Continuous

Hazard Warning:

- Operating Room Temperature: 25 ± 2 °C
- Testing Voltage: 13.5 V
- Testing Current: Rated Load (Hazard Warning)
- Test Duration: 100 H Continuous

TEST SETUP:





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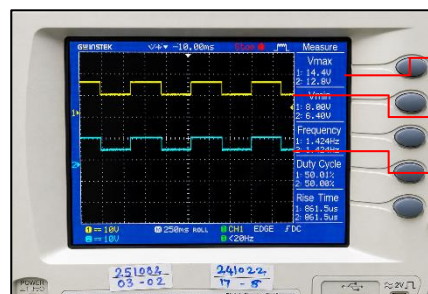
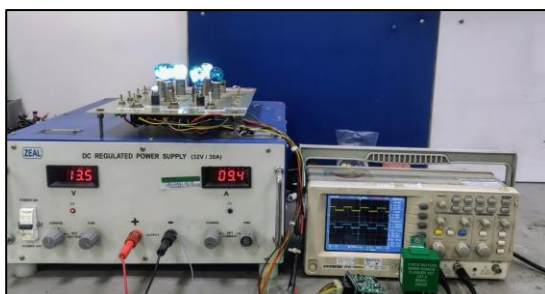
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TURN SIGNALS & HAZARD SIGNALS:

LEFT SIGNAL	RIGHT SIGNAL	HAZARD SIGNAL
		

WAVEFORM DATA:



INPUT VOLTAGE

LEFT LOAD

RIGHT LOAD

TEST OBSERVATION:

Description	Load	Observations(at 13.5 V)	Remark
	LED& CLUSTER	Sample No:01 &02	
Turn Signal RH	3 x 21 W Lamp + 1x 2W LED	85	OK
Turn Signal LH	3 x 21 W Lamp + 1x 2W LED	85	OK
Hazard Mode	6 x 21 W Lamp + 2x 2 W LED+	85	OK
Fault condition Any one 21W Lamp failure	2 x 21 W Lamp + 1x 2 W LED	170	OK
Duty cycles for RH&LH	3 x 21 W Lamp + 1x 2 W LED	50.01%	OK
Duty cycles for Hazard Mode	6 x 21 W Lamp + 1x 2 W LED	50%	OK

NOTE: The observed flashes are within the specifications & the samples met the performance requirements.

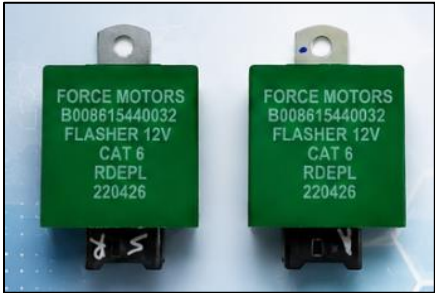
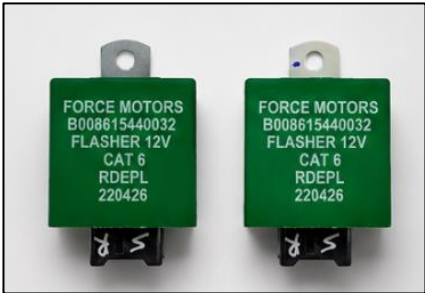


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TESTED SAMPLES PHOTOS:

BEFORE TEST SAMPLE – S:01 & 02	AFTER TEST SAMPLE – 01 & 02
	

OBSERVATION:

There is no significant change in the electrical performance of the flasher samples. The functionality remains satisfactory. The observed flashes are within the specified requirements & the samples met the performance requirements. All function of the Flasher performs as designed during & after the test (Class A).